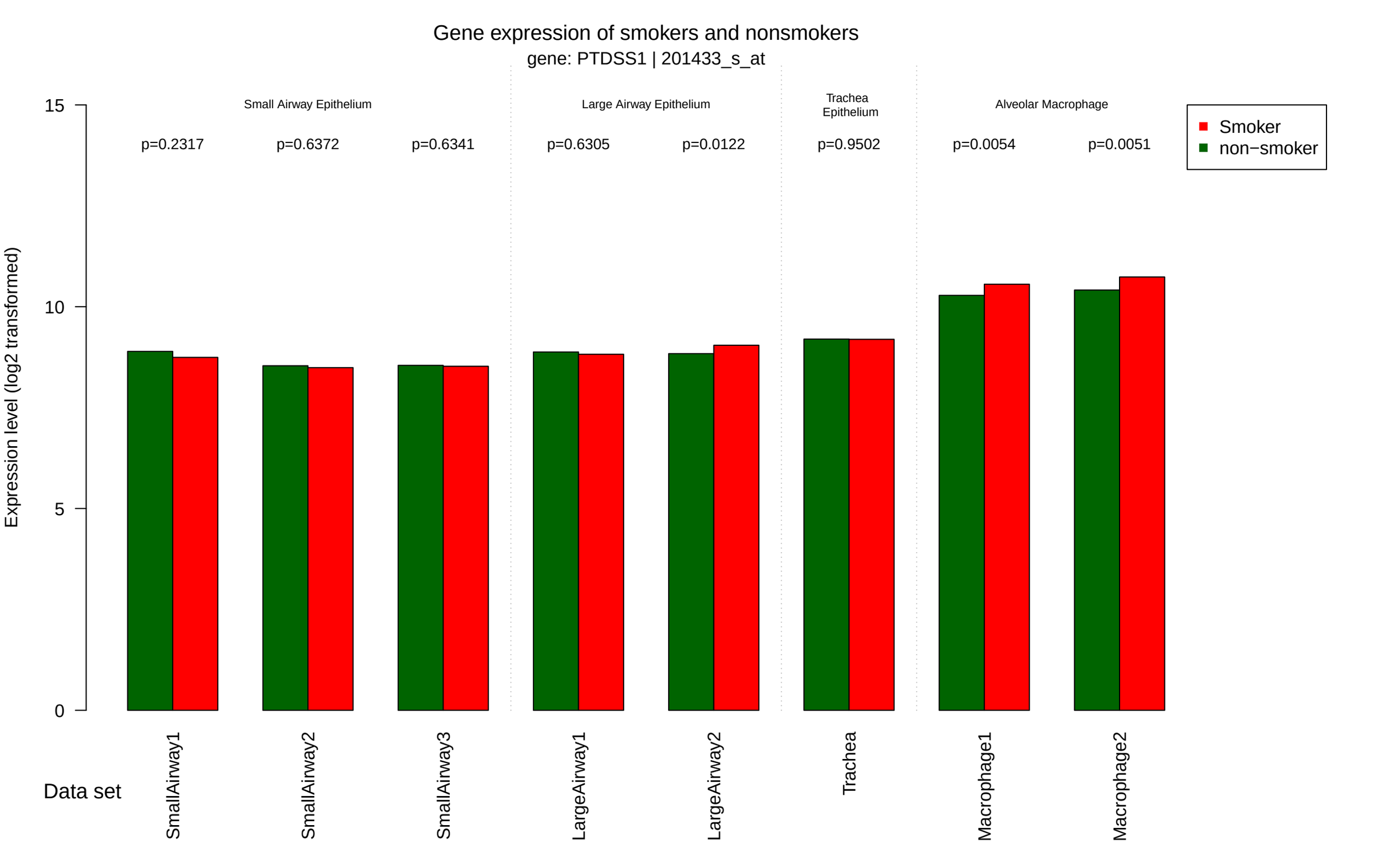
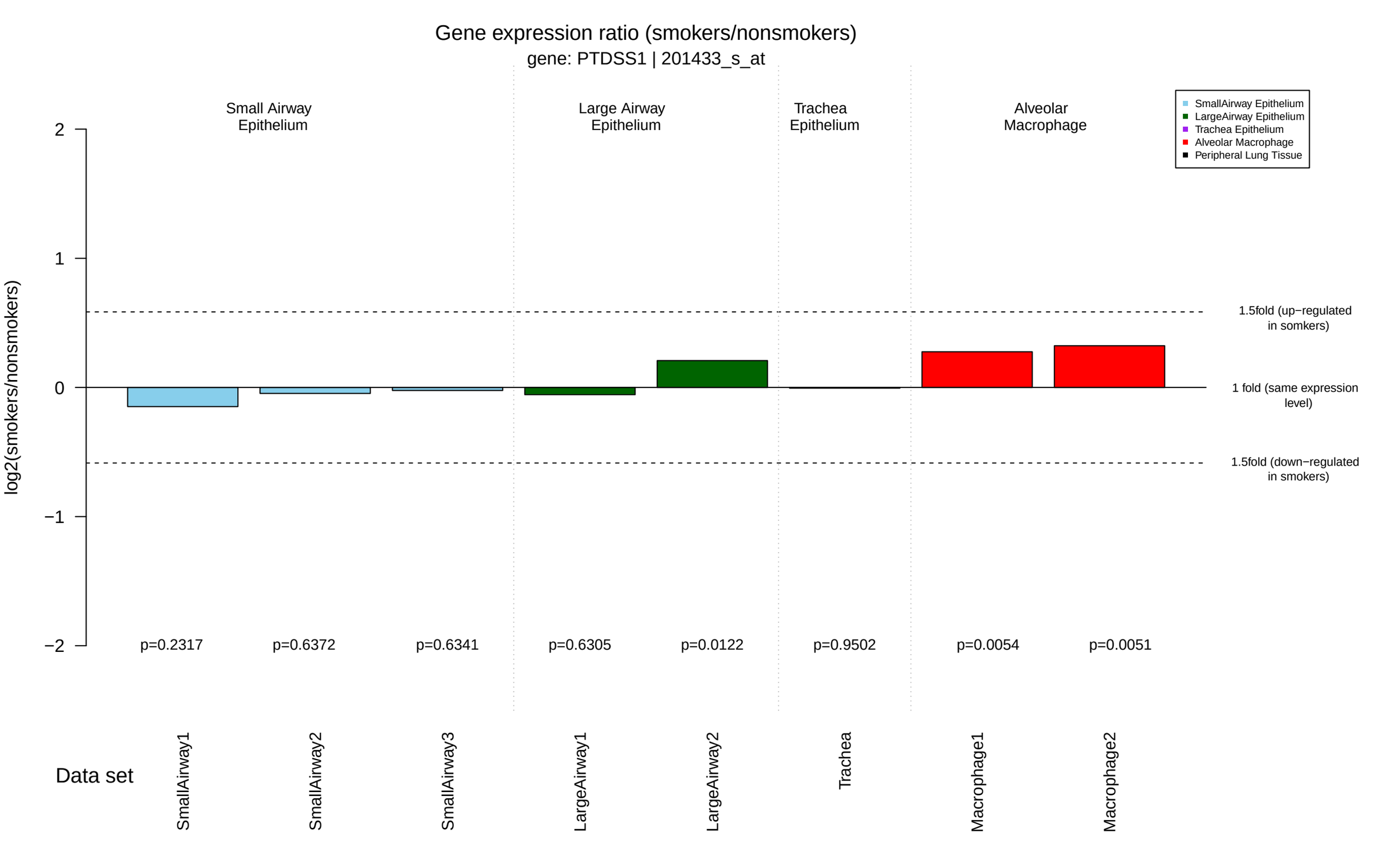
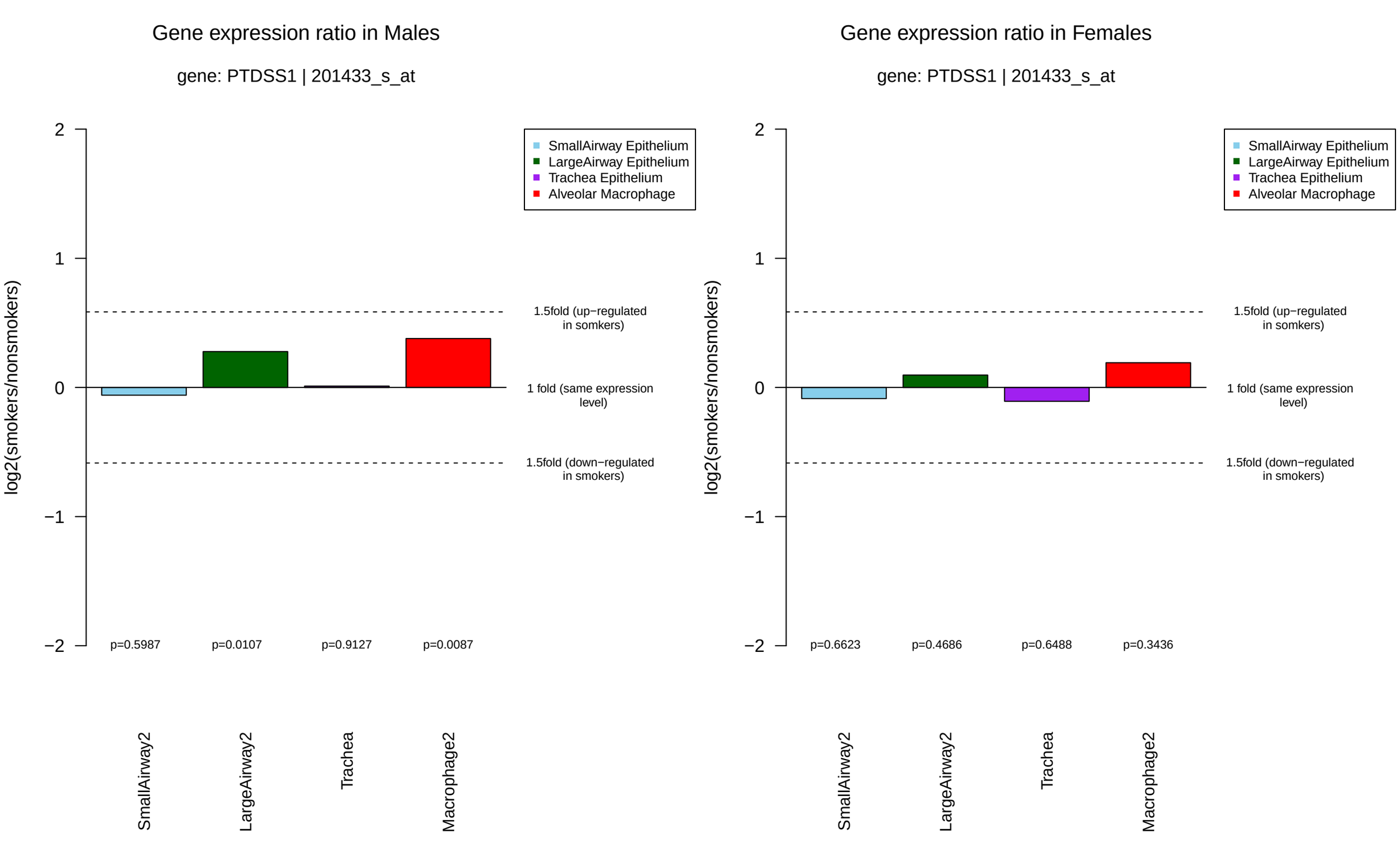
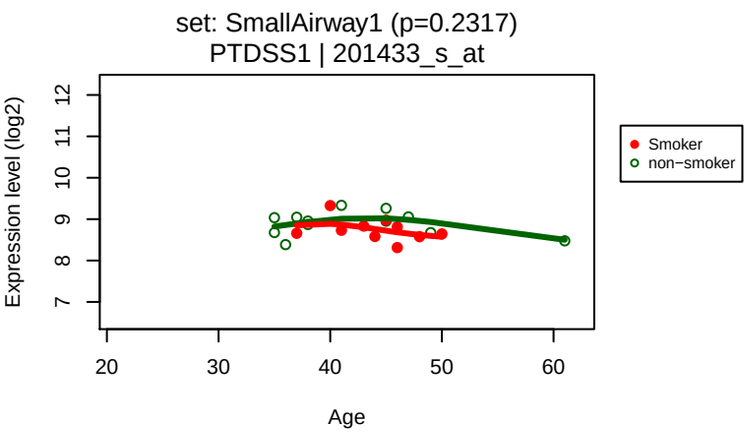
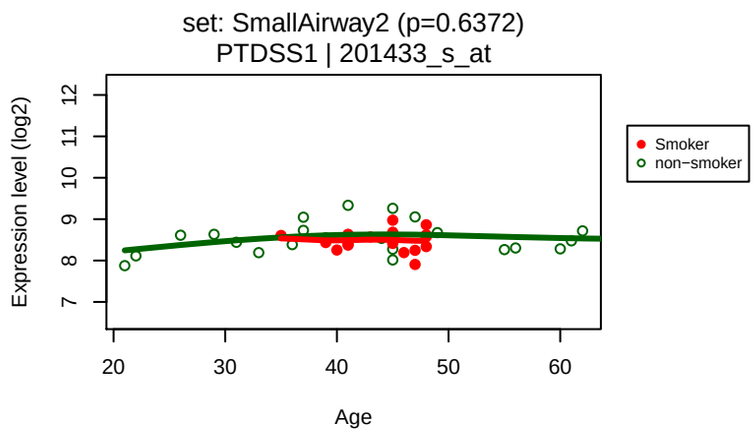
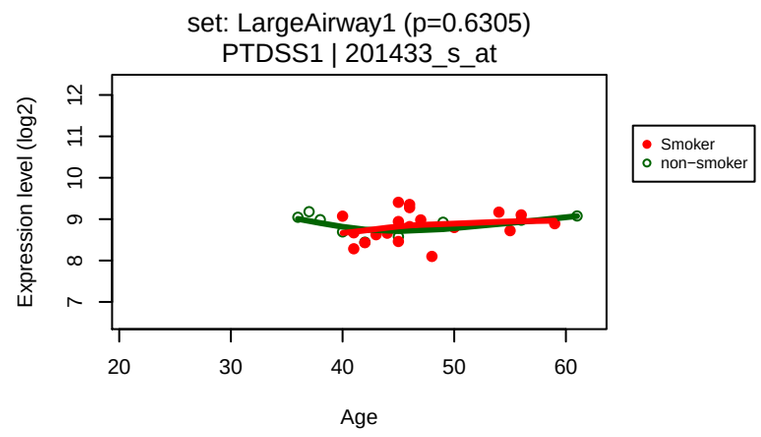

Gene expression of smokers and nonsmokers
gene: PTDSS1 | 201433_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2317
p=0.6372
p=0.6341
p=0.6305
p=0.0122
p=0.9502
p=0.0054
p=0.0051
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

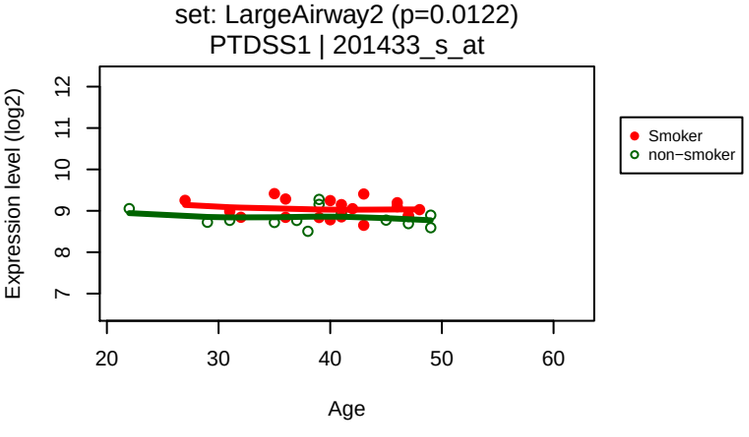

Gene expression ratio (smokers/nonsmokers)
gene: PTDSS1 | 201433_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2317
p=0.6372
p=0.6341
p=0.6305
p=0.0122
p=0.9502
p=0.0054
p=0.0051
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

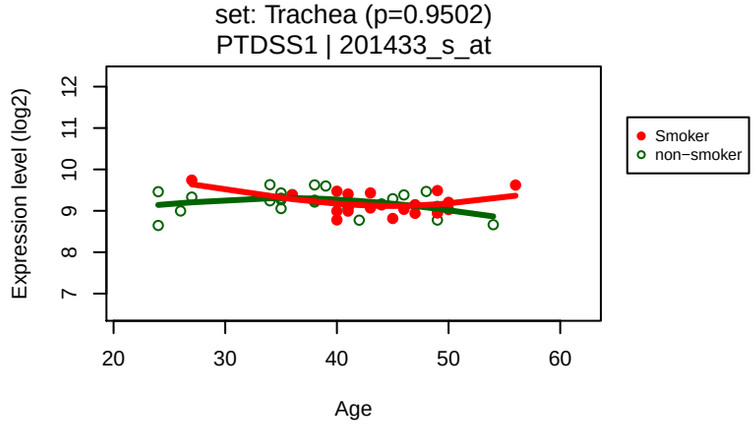

Gene expression ratio in Males
gene: PTDSS1 | 201433_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5987
p=0.0107
p=0.9127
p=0.0087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTDSS1 | 201433_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6623
p=0.4686
p=0.6488
p=0.3436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

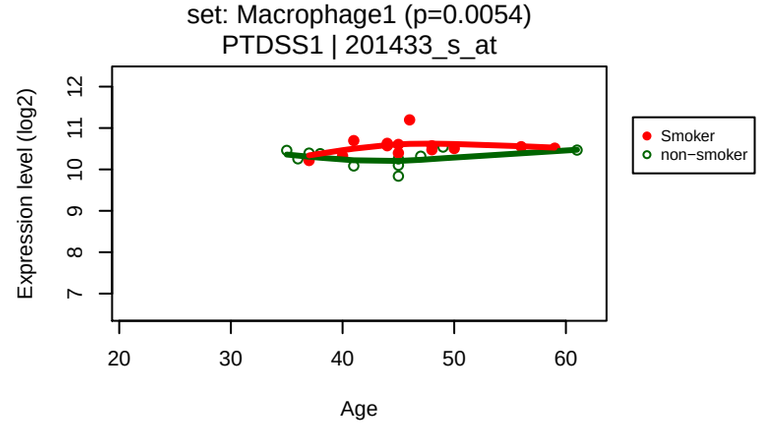

set: SmallAirway1 (p=0.2317)
PTDSS1 | 201433_s_at
Expression level (log2)
Age
Smoker
non−smoker

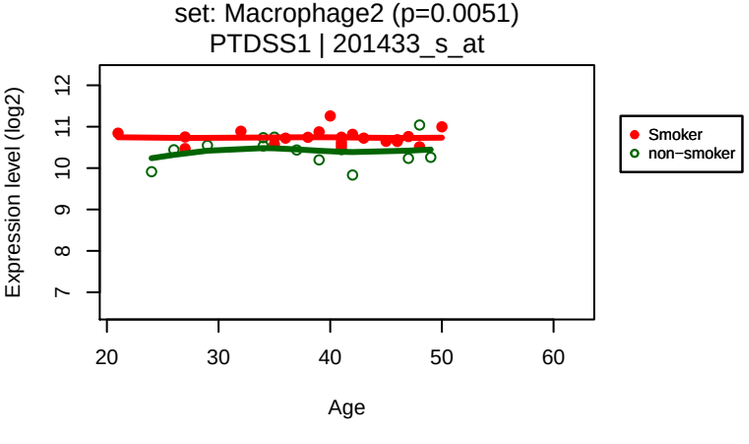

set: SmallAirway2 (p=0.6372)
PTDSS1 | 201433_s_at
Expression level (log2)
Age
Smoker
non−smoker

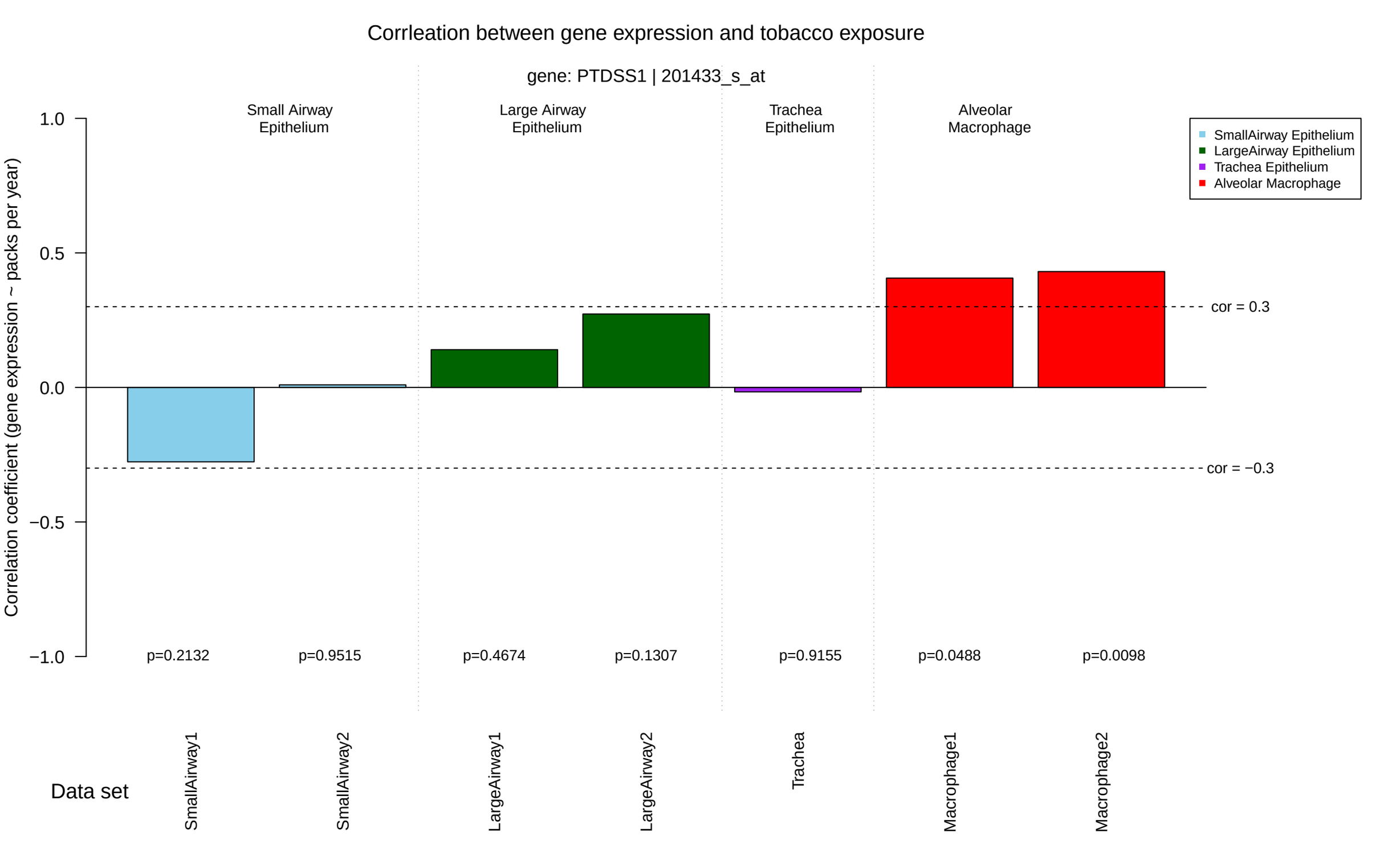

set: LargeAirway1 (p=0.6305)
PTDSS1 | 201433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0122)
PTDSS1 | 201433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9502)
PTDSS1 | 201433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0054)
PTDSS1 | 201433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0051)
PTDSS1 | 201433_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTDSS1 | 201433_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2132
p=0.9515
p=0.4674
p=0.1307
p=0.9155
p=0.0488
p=0.0098
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2